THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following correctly describes the live wire? 1

(a) At high potential with respect to earth **(b)** Has green insulation

(c) At zero potential **(d)** Connected to the metal body

2. Assertion (A): The fuse is connected in the live wire. Reason (R): It disconnects the appliance from the supply when excess current flows. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

3. Which of the following does not produce a magnetic field? 1

(a) A stationary charge **(b)** A current-carrying wire

(c) A moving charge **(d)** A bar magnet

4. A person touches a metal body of an appliance whose live wire is loose and touching the body. If the body is earthed 1

(a) the fuse blows and the person is safe **(b)** the appliance runs better

(c) there is no change **(d)** the person is electrocuted

5. Assertion (A): The direction of the force on a charge moving in a magnetic field is perpendicular to the velocity. Reason (R): The force cannot change the speed of the charged particle. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

6. An iron nail is placed in a solenoid carrying current and then the current is switched off. The nail 1

(a) remains strongly magnetised always

(b) loses most of the magnetism if it is soft iron

(c) becomes non-metal

(d) attracts everything

7. If the iron core is removed from an electromagnet, the strength 1

(a) increases **(b)** is unchanged **(c)** decreases **(d)** is infinite

8. Assertion (A): An electric motor has a split-ring commutator. Reason (R): The commutator reverses the direction of current in the coil every half rotation so that the coil keeps turning the same way. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. **Assertion (A):** The metallic body of an electrical appliance is connected to the earth wire. 1
Reason (R): The earth wire provides a low-resistance path for leakage current and prevents electric shocks.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A current-carrying conductor placed perpendicular to a magnetic field experiences a force. 1
Reason (R): The direction of the force reverses when the direction of the current is reversed.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two solenoids A and B have the same current and length, but A has 200 turns and B has 400 turns. Compare the fields. 2

12. Why does a current-carrying coil suspended freely in the earth's field rest along the north-south direction? 2

13. Explain why the iron filings near the poles of a bar magnet are crowded. 2

14. If a bar magnet is cut into two pieces along its length, what are the poles of each piece? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a generator the armature rotates at 50 revolutions per second. State the frequency of the AC produced and its period. 3

16. In a closed circuit with a galvanometer and a coil, a magnet falls through the coil. Describe the sign of the galvanometer deflection when it enters and leaves. 3

17. If the earth had no magnetic field, why would a compass needle be of no use? 3

OR

What is meant by earthing? Explain with an example why it is necessary for a refrigerator. 3

18. Give reasons: 3
(a) why the metal body is earthed
(b) why the fuse is in the live wire
(c) why a thick wire is used in the 15 A circuit.

OR

What would happen to an appliance if the neutral wire in a 220 V circuit is broken? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mr Gupta is installing a 2 kW geyser at his house in Lucknow, where the supply is 220 V. The electrician connects the geyser through a separate 15 A circuit with a fuse and an earth wire. 5
(i) What is the colour of the live wire and the earth wire as per the convention?
(ii) Calculate the current drawn by the geyser.
(iii) Why is the geyser not connected to the 5 A lighting circuit?
(iv) State the function of the fuse and of the earth wire.

20. Explain the factors on which the force on a current-carrying conductor in a field depends and how the direction can be reversed. A rod carries current across a uniform field. State the direction of the force with the left-hand rule if the field points upward along the page and the current flows to the right along the page. 5

OR

A coil is connected to a galvanometer. A magnet is pushed into it, held for a few seconds and then pulled out. Describe the galvanometer readings in the three stages and explain. 5

END OF QUESTION PAPER — Total: 40 Marks